

Worksheet 102 · Spot on facts, pp. 226–227

# Technologies that change society

The key facts at a glance — the connections are yours to make. Your basis for the Kolloquium.

## 1 Developments in the 21st century

### Genetic engineering

CRISPR edits the genetic code precisely.

+ drought-resistant crops, new treatments

– morality; interference with DNA

### Eco-technology

Solar, hydrogen and wind power are cheaper; more e-cars.

+ less fossil fuel, ocean clean-ups

– raw materials (lithium) harm the environment

### AI & digital content

Prompts to large language models create texts, media, programs.

+ information re-used in myriad contexts

– where does content come from — can we trust it?

Connect: All three bring a hope and a new problem. Which development shows this most clearly — and why?

## 2 Artificial intelligence (AI)

AI = the simulation of human intelligence in machines: learning, problem-solving, reasoning, perceiving.

### Weak / narrow AI

single tasks, e.g. personal assistants, video games

### Strong / general AI

complex, human-like, e.g. (book) self-driving cars, robots\*

### Impact

+ everyday life, education, health

– jobs, deepfakes, misinformation, copyright

\* Most experts today call self-driving cars narrow AI; general AI means a still hypothetical human-level intelligence.

Connect: Why does checking data matter more than ever now that AI creates content?

## 3 Augmented reality (AR) and virtual reality (VR) — both extended reality (XR)

### AR

What it does adds digital overlays to the real world

Devices phone, tablet (camera)

Used in entertainment, medicine, shopping

### VR

replaces the real world with a 360° environment

headset, motion controllers

gaming, tourism, education, medicine, warfare

Connect: How are AR, VR and AI related? Name one example you know (e.g. Virtual Helsinki, p. 230).

## 4 Utopia or dystopia — where will technology take us?

**Utopia** ou-topos = “no place”, eu-topos = “good place” (Thomas More, 1516).

An ideal society that criticises the present by showing a better world.

e.g. Butler, Erewhon (1872); Bellamy, Looking Backward (1888)

**Dystopia** dys-topos = “bad place”.

A worst-case society, often in the near future, that warns about contemporary problems.

e.g. Brave New World (1932), Nineteen Eighty-Four (1949), The Hunger Games (2008)

Connect: Which technology from boxes 1–3 could be the starting point of a dystopia? Describe that society in one or two sentences.

# Kolloquium Carousel

In Part 2 of the Kolloquium you answer questions on topics from other semesters — no notes. Train it now.

## How it works

- 1 Double circle: inside candidates, outside examiners.
- 2 Examiner reads a card — 15 s to think, 2 min to answer.
- 3 Follow-up question (1 min), then rubric + two stars and a wish.
- 4 Outer circle moves on. Swap roles after two rounds.

## Answer like a candidate

Define → Explain → Example → Evaluate

## Question cards (follow-up in italics)

### 1 AR vs VR

Explain the difference between AR and VR.

*Follow-up:* Which will change everyday life more?

### 2 AI

What is AI? Weak vs strong AI.

*Follow-up:* Where does AI help — where does it worry you?

### 3 Trust

Why is it harder to know where online content comes from?

*Follow-up:* How can you protect yourself?

### 4 Genes

What does CRISPR make possible?

*Follow-up:* Where would you draw the line?

### 5 Terms

Where do utopia and dystopia come from?

*Follow-up:* Why are dystopias so popular today?

### 6 Big picture

Which technology is the most life-changing?

*Follow-up:* How will it affect your children's lives?

## Examiner's sheet 0 = not yet · 1 = partly · 2 = mostly · 3 = fully

|                                         | Round 1 | Round 2 |
|-----------------------------------------|---------|---------|
| Candidate                               |         |         |
| Content — accurate facts and terms      | /3      | /3      |
| Structure — clear line of thought       | /3      | /3      |
| Language — topic words, range, accuracy | /3      | /3      |
| Interaction — reacts to the follow-up   | /3      | /3      |
| Total                                   | /12     | /12     |

**After the carousel — your Kolloquium note** Which card was hardest? What would you need to answer it well next time?

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